

Stat 286: Causal Inference with Applications, Fall 2025

Section 10: Weighting

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1 Weighting

1.1 Motivation

In the previous module we have encountered the matching estimators (minus the bias correction)

$$\hat{\tau}_{\text{ATT}}^{\text{m}} = \frac{1}{n_1} \sum_{i=1}^n T_i \left(Y_i - \frac{1}{M_i} \sum_{j \in \mathcal{M}_i} Y_j \right),$$

$$\hat{\tau}_{\text{ATE}}^{\text{m}} = \frac{1}{n} \sum_{i=1}^n \left(T_i Y_i + (1 - T_i) \frac{1}{M_i} \sum_{j \in \mathcal{M}_i} Y_j \right) - \left((1 - T_i) Y_i + T_i \frac{1}{M_i} \sum_{j \in \mathcal{M}_i} Y_j \right),$$

with appropriately defined matching set $\mathcal{M}_i = \mathcal{M}_i(\mathbf{X}_1, \dots, \mathbf{X}_n, T_1, \dots, T_n)$. We could have equivalently written these as

$$\hat{\tau}_{\text{ATT}}^{\text{m}} = \frac{1}{n_1} \sum_{i=1}^n T_i Y_i - \frac{1}{n_1} \sum_{i=1}^n (1 - T_i) w_i^{\text{ATT}} Y_i,$$

$$\hat{\tau}_{\text{ATE}}^{\text{m}} = \frac{1}{n} \sum_{i=1}^n (2T_i - 1) w_i^{\text{ATE}} Y_i.$$

with weights $w_i^{\text{ATT}} = \sum_{j=1}^n \frac{1}{M_j} \mathbf{1}\{i \in \mathcal{M}_j\}$ and $w_i^{\text{ATE}} = 1 + w_i^{\text{ATT}}$. This motivates considering *weighting* estimators of the form

$$\hat{\tau} = \frac{1}{n} \sum_{i=1}^n T_i w_{1i} Y_i - \frac{1}{n} \sum_{i=1}^n (1 - T_i) w_{0i} Y_i, \quad \text{or simply, } \hat{\tau} = \frac{1}{n} \sum_{i=1}^n w_i Y_i,$$

and studying their weights directly, without concerning ourselves with the (often cumbersome) matching procedure. In this module, we will discuss two approaches to weighting: *Propensity Score Weighting* and *Double Robust Estimation*.

2 Propensity Score Weighting

2.1 The Central Role of Propensity Score

This subsection is taken from Ding [2023, Section 11]. Assuming i.i.d. sampling and *SUTVA*, in observational causal inference we have four random variables associated with each unit: $\{\mathbf{X}, T, Y(0), Y(1)\}$. Its joint distribution can be factorized

$$\Pr(\mathbf{X}, T, Y(0), Y(1)) = \Pr(\mathbf{X}) \Pr(Y(0), Y(1) \mid \mathbf{X}) \Pr(T \mid \mathbf{X}, Y(0), Y(1)),$$

where

- $\Pr(\mathbf{X})$ is the covariate distribution,
- $\Pr(Y(0), Y(1) \mid \mathbf{X})$ is the outcome distribution conditional on the covariates $\mathbf{X}$,
- $\Pr(T \mid \mathbf{X}, Y(0), Y(1))$ is the treatment distribution conditional on $\{\mathbf{X}, Y(0), Y(1)\}$.

If we knew the joint distribution we could identify *any* causal estimand. However, apriori, only $\Pr(\mathbf{X})$ is identifiable as the other two building blocks depend on the causal quantities $\{Y(0), Y(1)\}$.

In this section, we focus on the conditional treatment distribution and redefine the propensity score as

$$e(\mathbf{X}, Y(0), Y(1)) = \Pr(T = 1 \mid \mathbf{X}, Y(0), Y(1)).$$

Under strong ignorability, $T \perp\!\!\!\perp \{Y(0), Y(1)\} \mid \mathbf{X}$, this reduces to the familiar

$$e(\mathbf{X}) = \Pr(T = 1 \mid \mathbf{X}).$$

The propensity score was first introduced as a key concept in Rosenbaum and Rubin [1983] and many of the following results have been adapted from there.

Theorem 1 (Propensity score as one-dimensional summary). *If $T \perp\!\!\!\perp \{Y(0), Y(1)\} \mid \mathbf{X}$ then $T \perp\!\!\!\perp \{Y(0), Y(1)\} \mid e(\mathbf{X})$.*

Theorem 1 states that if strong ignorability holds conditional on covariates $\mathbf{X}$ then it is enough adjust for the propensity score $e(\mathbf{X})$. This is quite remarkable as $\mathbf{X}$ can be very high-dimensional, but $e(\mathbf{X}) \in [0, 1]$. Therefore we can view the propensity score as a dimension reduction tool.

Proof. By conditional independence we need to show that

$$\Pr(T = 1 \mid e(\mathbf{X}), Y(0), Y(1)) = \Pr(T = 1 \mid e(\mathbf{X})).$$

We have

$$\begin{aligned} \Pr(T = 1 \mid e(\mathbf{X}), Y(0), Y(1)) &= \mathbb{E}[T \mid e(\mathbf{X}), Y(0), Y(1)] \\ &= \mathbb{E}[\mathbb{E}[T \mid \mathbf{X}, e(\mathbf{X}), Y(0), Y(1)] \mid e(\mathbf{X}), Y(0), Y(1)] \quad (\text{LIE}) \\ &= \mathbb{E}[\mathbb{E}[T \mid \mathbf{X}, Y(0), Y(1)] \mid e(\mathbf{X}), Y(0), Y(1)] \quad (e(\mathbf{X}) \text{ is a function of } \mathbf{X}) \\ &= \mathbb{E}[\mathbb{E}[T \mid \mathbf{X}] \mid e(\mathbf{X}), Y(0), Y(1)] \quad (\text{unconfoundedness}) \\ &= \mathbb{E}[\Pr(T = 1 \mid \mathbf{X}) \mid e(\mathbf{X}), Y(0), Y(1)] \\ &= \mathbb{E}[e(\mathbf{X}) \mid e(\mathbf{X}), Y(0), Y(1)] \\ &= e(\mathbf{X}) \\ &= \mathbb{E}[e(\mathbf{X}) \mid e(\mathbf{X})] \\ &= \mathbb{E}[\mathbb{E}[T \mid \mathbf{X}] \mid e(\mathbf{X})] \\ &= \mathbb{E}[\mathbb{E}[T \mid e(\mathbf{X}), \mathbf{X}] \mid e(\mathbf{X})] \quad (e(\mathbf{X}) \text{ is a function of } \mathbf{X}) \\ &= \mathbb{E}[T \mid e(\mathbf{X})] \quad (\text{reverse LIE}) \\ &= \Pr(T = 1 \mid e(\mathbf{X})). \end{aligned}$$

□

Remark 1. Theorem 1 also suggests that if we discretize the propensity score into K quantiles $e_1, \dots, e_K$ that unconfoundedness

$$T_i \perp\!\!\!\perp \{Y_i(0), Y_i(1)\} \mid e(\mathbf{X}_i) = e_k, \quad \text{for } k = 1, \dots, K.$$

holds approximately for each quantile. We can use this to estimate the ATE. Following Wager [2024, Section 2] we can implement this by,

1. Estimate $\hat{e}(\mathbf{x})$,
2. Sort the estimated propensity score: $\hat{e}(\mathbf{X}_{i_1}) \leq \hat{e}(\mathbf{X}_{i_2}) \leq \dots \leq \hat{e}(\mathbf{X}_{i_n})$,
3. Split the sample into K evenly sized strata using the sorted propensity score,
4. For each stratum $k = 1, \dots, K$ estimate

$$\hat{\tau}_k = \frac{1}{|\{i : T_i = 1, e(\mathbf{X}_i) = e_k\}|} \sum_{i: e(\mathbf{X}_i) = e_k} T_i Y_i - \frac{1}{|\{i : T_i = 0, e(\mathbf{X}_i) = e_k\}|} \sum_{i: e(\mathbf{X}_i) = e_k} (1 - T_i) Y_i,$$

5. Combine: $\hat{\tau}^{\text{ATE}} = \frac{1}{K} \sum_{k=1}^K \hat{\tau}_k$.

For the next property, we need the following definition

Definition 1 (Balancing Score). A function of the covariates $b(\mathbf{X})$ is a balancing score if

$$T_i \perp\!\!\!\perp \mathbf{X}_i \mid b(\mathbf{X}_i).$$

A trivial balancing score is $b(\mathbf{x}) = \mathbf{x}$. We are most interested in lower dimensional balancing scores.

Lemma 1 (Imbens and Rubin [2015] Lemma 12.2, Balancing Property of the Propensity Score). *The propensity score $e(\mathbf{X})$ is a balancing score, i.e., $T_i \perp\!\!\!\perp \mathbf{X}_i \mid e(\mathbf{X}_i)$.*

It turns out that for every balancing score $b(\mathbf{x})$ we have

$$T_i \perp\!\!\!\perp \{Y_i(0), Y_i(1)\} \mid b(\mathbf{X}_i),$$

so the propensity score is not that special in that perspective. What makes the propensity score special is that it is the *coarsest* balancing score.

Lemma 2 (Imbens and Rubin [2015] Lemma 12.3, Coarseness of Balancing Scores). *The propensity score is the coarsest balancing score. That is, for every balancing score b there exists a function f_b such that $e(\mathbf{x}) = f_b(b(\mathbf{x}))$.*

Since the propensity score is the coarsest possible balancing score, it *maximally* compresses all the information given in the covariates $\mathbf{X}$ needed to adjust for.

2.2 Inverse Probability Weighting

How does this connects to weighting?

Theorem 2. *Assuming i.i.d. sampling, SUTVA, Unconfoundedness and Overlap we have*

$$\mathbb{E}[Y(1)] = \mathbb{E}\left[\frac{TY}{e(\mathbf{X})}\right], \quad \mathbb{E}[Y(0)] = \mathbb{E}\left[\frac{(1-T)Y}{1-e(\mathbf{X})}\right],$$

and

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}\left[\frac{TY}{e(\mathbf{X})} - \frac{(1-T)Y}{1-e(\mathbf{X})}\right].$$

Proof.

$$\begin{aligned} \mathbb{E}\left[\frac{TY}{e(\mathbf{X})}\right] &= \mathbb{E}\left[\frac{TY(1)}{e(\mathbf{X})}\right] \\ &= \mathbb{E}\left[\mathbb{E}\left[\frac{TY(1)}{e(\mathbf{X})} \mid \mathbf{X}\right]\right] && \text{(LIE)} \\ &= \mathbb{E}\left[\frac{\mathbb{E}[Y(1) \mid \mathbf{X}]}{e(\mathbf{X})} \mathbb{E}[T \mid \mathbf{X}]\right] && \text{(Unconfoundedness)} \\ &= \mathbb{E}\left[\frac{\mathbb{E}[Y(1) \mid \mathbf{X}]}{e(\mathbf{X})} e(\mathbf{X})\right] \\ &= \mathbb{E}[Y(1)]. \end{aligned}$$

A similar argument can be applied to $\mathbb{E}[Y(0)]$. Taking the difference yields the claim. $\square$

Theorem 2 suggest the following weighting estimator for the ATE

$$\hat{\tau}_{\text{ATE}}^{\text{ht}} = \frac{1}{n} \sum_{i=1}^n \frac{T_i Y_i}{\hat{e}(\mathbf{X}_i)} - \frac{1}{n} \sum_{i=1}^n \frac{(1-T_i) Y_i}{1-\hat{e}(\mathbf{X}_i)},$$

where $\hat{e}(\mathbf{X}_i)$ is the estimated propensity score. This estimator is called the inverse propensity score weighting (IPW) estimator, sometimes also called Horvitz–Thompson (HT) estimator, named after Horvitz and Thompson [1952], who proposed it first. In theory $\hat{\tau}_{\text{ATE}}^{\text{ht}}$ is an unbiased estimator if the propensity score is estimated correctly.

Intuition. Consider the treated term $\frac{1}{n} \sum_{i=1}^n \frac{T_i Y_i}{\hat{e}(\mathbf{X}_i)}$. Each treated outcome is divided by its treatment probability, so rarely treated covariate profiles ($\hat{e}_i$ small) receive larger weights and stand in for many similar units who were not treated due to their low $\hat{e}_i$. This reweights the population to what would happen if everyone were treated. A similar argument can be applied to the second term.

Challenges. There are two main issues with $\hat{\tau}_{\text{ATE}}^{\text{HT}}$.

1. $\hat{e}(\mathbf{X}_i)$ can be very large or small. This can lead to numerical instability but also to very large weights which dominate the whole estimator. We essentially then only compare those units. This is especially bad if our scores are not estimated well.
2. The estimator is not shift invariant, i.e., for $Y'_i = Y_i + c$ we obtain a different estimator, although the ATE stays the same.

We try to address these by introducing the Hajek estimator [Hájek, 1971]

$$\hat{\tau}_{\text{ATE}}^{\text{hajek}} = \frac{\sum_{i=1}^n \frac{T_i Y_i}{\hat{e}(\mathbf{X}_i)}}{\sum_{i=1}^n \frac{T_i}{\hat{e}(\mathbf{X}_i)}} - \frac{\sum_{i=1}^n \frac{(1-T_i) Y_i}{1-\hat{e}(\mathbf{X}_i)}}{\sum_{i=1}^n \frac{1-T_i}{1-\hat{e}(\mathbf{X}_i)}}.$$

The Hajek estimator has normalized weights but is not unbiased (even theoretically). Often desired theoretical properties are proved with $\hat{\tau}_{\text{ATE}}^{\text{ht}}$ but for practical considerations $\hat{\tau}_{\text{ATE}}^{\text{hajek}}$ is used.

Remark 2. Define the so called *clever covariate*

$$H = H(T, \mathbf{X}) = \frac{T}{e(\mathbf{X})} - \frac{1-T}{1-e(\mathbf{X})},$$

such that

$$\text{ATE} = \mathbb{E}[YH] \approx \frac{1}{n} \sum_{i=1}^n Y_i \hat{H}_i.$$

This emphasizes the role of the IPW as a weighting estimator, but also serves other purposes. By definition $\mathbb{E}[H | \mathbf{X}] = 0$. We can use this as a diagnostic tool by regressing $\hat{H} \sim \mathbf{X}$ and observing whether $\mathbf{X}$ is predictive of H (it should not be!). If it is, covariate balance is violated. Furthermore, we can define the augmented outcome YH and regress $YH \sim \mathbf{X}$. This provides us with an estimate of the CATE $\mathbb{E}[Y(1) - Y(0) | \mathbf{X}]$.

Before we leave we quickly mention that estimated propensity scores can be very close to zero or one, blowing up the estimator. Authors have suggested truncating the propensity score by

$$\hat{e}_i^{\text{trun}}(\mathbf{X}_i) = \max[\alpha_L, \min\{\hat{e}_i(\mathbf{X}_i), \alpha_U\}],$$

where, for example, $\alpha_L = 0.05$ and $\alpha_U = 0.95$. This stabilizes the estimator but also changes the estimand in an arbitrary way. For a more thorough discussion we refer to Ding [2023, Section 11.2.3].

2.3 Estimating the Propensity Score

Even though the propensity score has very desirable properties, the estimated propensity score might not. Thus, it is critical to estimate it as good as possible. In this section we follow Imai and Ratkovic [2014].

Typically researcher assume a parametric model on the propensity score $e(\mathbf{X}_i) = \pi_\theta(\mathbf{X}_i)$ where $\theta \in \Theta$. A popular choice is logistic regression

$$\pi_\theta(\mathbf{X}_i) = \frac{\exp(\mathbf{X}_i^\top \theta)}{1 + \exp(\mathbf{X}_i^\top \theta)}.$$

Often θ is estimated via the maximum likelihood approach, i.e., maximizing the log-likelihood $\hat{\theta} = \arg \max_{\theta} l(\theta)$ with

$$l(\theta) = \arg \max_{\theta} \sum_{i=1}^n T_i \log(\pi_\theta(\mathbf{X}_i)) - (1 - T_i) \log(1 - \pi_\theta(\mathbf{X}_i)).$$

Equivalently we could solve for θ in

$$\frac{\partial}{\partial \theta} l(\theta) = \frac{1}{n} \sum_{i=1}^n \left(\frac{T_i}{\pi_\theta(\mathbf{X}_i)} - \frac{1 - T_i}{1 - \pi_\theta(\mathbf{X}_i)} \right) \pi'_\theta(\mathbf{X}_i) = 0, \quad (1)$$

where $\pi'_\theta(\mathbf{X}_i) = \frac{\partial}{\partial \theta} \pi_\theta(\mathbf{X}_i)$. Assuming for a moment that $\pi_\theta(\mathbf{X}_i) = e(\mathbf{X}_i)$ holds then the population version of the above condition yields

$$\mathbb{E} \left[\frac{T_i}{\pi_\theta(\mathbf{X}_i)} \pi'_\theta(\mathbf{X}_i) \right] = \mathbb{E} \left[\frac{1 - T_i}{1 - \pi_\theta(\mathbf{X}_i)} \pi'_\theta(\mathbf{X}_i) \right] = \mathbb{E}[\pi'_\theta(\mathbf{X}_i)],$$

where the last equation follows from the balancing properties of the propensity score. Thus, Equation (1) can be interpreted as *balancing* $\pi'_\theta(\mathbf{X}_i)$.

Instead of balancing $\pi'_\theta(\mathbf{X}_i)$ which feels arbitrary Imai and Ratkovic [2014] propose to balance for a general function $f(\mathbf{X})$,

$$\mathbb{E} \left[\frac{T_i}{\pi_\theta(\mathbf{X}_i)} \pi'_\theta(\mathbf{X}_i) \right] = \mathbb{E} \left[\frac{1 - T_i}{1 - \pi_\theta(\mathbf{X}_i)} \pi'_\theta(\mathbf{X}_i) \right] = \mathbb{E}[f(\mathbf{X}_i)],$$

and estimating this via solving the sample analogue

$$\frac{1}{n} \sum_{i=1}^n \left(\frac{T_i}{\pi_\theta(\mathbf{X}_i)} - \frac{1 - T_i}{1 - \pi_\theta(\mathbf{X}_i)} \right) f(\mathbf{X}_i) = 0.$$

Typical choices include $f(\mathbf{X}_i) = [\mathbf{X}_i, \mathbf{X}_i^2, \dots, \mathbf{X}_i^k]$, i.e., balancing the first k moments. This is motivated by the fact that the moments determine the moment generating function and thus the distribution. In practice, balancing all moments is infeasible (infinity many, even for $k > 1$ overdetermined system) and so searchers typically set $k = 1, 2$. The authors call this procedure covariate balancing propensity score or CBPS for short.

However, this begs the question, what is the optimal f to balance for? In Fan et al. [2022] the authors show that the optimal f is given by

$$f(\mathbf{X}_i) = e(\mathbf{X}_i)\mu_0(\mathbf{X}_i) + (1 - e(\mathbf{X}_i))\mu_1(\mathbf{X}_i),$$

where $\mu_t(\mathbf{x}) = \mathbb{E}[Y \mid T = t, \mathbf{X} = \mathbf{x}]$ is the outcome function. Unfortunately, f depends on the unknown propensity score (which we want to estimate in the first place) and the unknown outcome function. Fan et al. [2022] propose an estimation procedure circumventing this problem based on method of moments, called optimal CBPS (oCBPS).

2.4 Asymptotic Normality

Theorem 3 (Fan et al. [2022] Theorem 3.2, Asymptotic Normality of the IPW via oCBPS). *Assume that the propensity score is estimated via oCBPS. Under i.i.d. sampling, SUTVA, Unconfoundedness and Overlap we have*

$$\sqrt{n}(\hat{\tau}_{\text{ATE}}^{\text{ht}} - \text{ATE}) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

The limiting variance is optimal, see Fan et al. [2022] for details.

We emphasize that other estimating procedures also yield a central limit theorem. Often we simply estimate the variance and confidence interval via bootstrapping.

3 Double Robust Estimation

3.1 Constructing the Estimator

So far we have seen two distinct ways to characterize and estimate the ATE under the usual assumptions.

Regression Approach.

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y \mid T = 1, \mathbf{X}] - \mathbb{E}[Y \mid T = 0, \mathbf{X}] = \mathbb{E}[\mu_1(\mathbf{X})] - \mathbb{E}[\mu_0(\mathbf{X})].$$

This gives rise to the regression estimator

$$\hat{\tau}_{\text{ATE}}^{\text{reg}} = \frac{1}{n} \sum_{i=1}^n \hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i).$$

This requires estimating $\hat{\mu}_t(\mathbf{x})$.

Weighting Approach.

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E} \left[\frac{TY}{e(\mathbf{X})} - \frac{(1-T)Y}{1-e(\mathbf{X})} \right],$$

with estimator

$$\hat{\tau}_{\text{ATE}}^{\text{ht}} = \frac{1}{n} \sum_{i=1}^n \frac{T_i Y_i}{\hat{e}(\mathbf{X}_i)} - \frac{1}{n} \sum_{i=1}^n \frac{(1-T_i) Y_i}{1-\hat{e}(\mathbf{X}_i)},$$

which requires estimating $\hat{e}(\mathbf{x})$.

It is natural to ask whether we can combine both strategies. This is indeed the case and gives rise to the augmented inverse probability weighting (AIPW) estimator [Robins et al., 1994]

$$\begin{aligned} \hat{\tau}_{\text{ATE}}^{\text{AIPW}} &= \frac{1}{n} \sum_{i=1}^n \left(\hat{\mu}_1(\mathbf{X}_i) + \frac{T_i(Y_i - \hat{\mu}_1(\mathbf{x}))}{\hat{e}(\mathbf{X}_i)} \right) - \frac{1}{n} \sum_{i=1}^n \left(\hat{\mu}_0(\mathbf{X}_i) + \frac{(1-T_i)(Y_i - \hat{\mu}_0(\mathbf{x}))}{1-\hat{e}(\mathbf{X}_i)} \right) \\ &= \underbrace{\frac{1}{n} \sum_{i=1}^n (\hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i))}_{\hat{\tau}_{\text{ATE}}^{\text{reg}}} + \underbrace{\frac{1}{n} \sum_{i=1}^n \left(\frac{T_i(Y_i - \hat{\mu}_1(\mathbf{X}_i))}{\hat{e}(\mathbf{X}_i)} - \frac{(1-T_i)(Y_i - \hat{\mu}_0(\mathbf{X}_i))}{1-\hat{e}(\mathbf{X}_i)} \right)}_{\approx \text{mean-zero noise}} \\ &= \underbrace{\frac{1}{n} \sum_{i=1}^n \left(\frac{T_i Y_i}{\hat{e}(\mathbf{X}_i)} - \frac{(1-T_i) Y_i}{1-\hat{e}(\mathbf{X}_i)} \right)}_{\hat{\tau}_{\text{ATE}}^{\text{ht}}} + \underbrace{\frac{1}{n} \sum_{i=1}^n \left(\hat{\mu}_1(\mathbf{X}_i) \left(1 - \frac{T_i}{\hat{e}(\mathbf{X}_i)} \right) - \hat{\mu}_0(\mathbf{X}_i) \left(1 - \frac{1-T_i}{1-\hat{e}(\mathbf{X}_i)} \right) \right)}_{\approx \text{mean-zero noise}} \end{aligned}$$

It is very easy to verify that if we replace $\hat{e}$ and $\hat{\mu}_t$ with their respective estimand that $\mathbb{E}[\hat{\tau}_{\text{ATE}}^{\text{AIPW}}] = \text{ATE}$. What is much more surprising is that we only need *one* of them to be correct for unbiasedness to hold. This is called *double robustness* and gives us two tries to estimate the correct model. The next result formalizes this claim.

Theorem 4 (Double Robustness of the AIPW). *Assuming i.i.d. sampling, SUTVA, Unconfoundedness and Overlap. If either the propensity score is correctly specified, i.e. $\hat{e}(\mathbf{x}) = e(\mathbf{x})$, or the outcome model is correctly specified, i.e., $\hat{\mu}_t = \mu_t$ for $t = 0, 1$, then*

$$\mathbb{E}[\hat{\tau}_{\text{ATE}}^{\text{AIPW}}] = \text{ATE}.$$

Thus if either model is consistent, $\hat{\tau}_{\text{ATE}}^{\text{AIPW}}$ is a consistent estimator.

Proof. We follow the proof of Ding [2023, Theorem 12.1]. We only show that

$$\mathbb{E} \left[\hat{\mu}_1(\mathbf{X}_i) + \frac{T_i(Y_i - \hat{\mu}_1(\mathbf{x}))}{\hat{e}(\mathbf{X}_i)} \right] = \mathbb{E}[Y_i(1)].$$

The proof for $\mathbb{E}[Y(0)]$ works similarly.

$$\begin{aligned}
& \mathbb{E} \left[\hat{\mu}_1(\mathbf{X}_i) + \frac{T_i(Y_i - \hat{\mu}_1(\mathbf{x}))}{\hat{e}(\mathbf{X}_i)} \right] - \mathbb{E}[Y(1)] \\
&= \mathbb{E} \left[\frac{T_i(Y_i(1) - \hat{\mu}_1(\mathbf{x}))}{\hat{e}(\mathbf{X}_i)} - (Y(1) - \hat{\mu}_1(\mathbf{X}_i)) \right] \\
&= \mathbb{E} \left[\left(\frac{T_i}{\hat{e}(\mathbf{X}_i)} - 1 \right) \times (Y_i(1) - \hat{\mu}_1(\mathbf{X}_i)) \right] \\
&= \mathbb{E} \left[\left(\frac{T_i - \hat{e}(\mathbf{X}_i)}{\hat{e}(\mathbf{X}_i)} \right) \times (Y_i(1) - \hat{\mu}_1(\mathbf{X}_i)) \right] \\
&= \mathbb{E} \left[\mathbb{E} \left[\left(\frac{T_i - \hat{e}(\mathbf{X}_i)}{\hat{e}(\mathbf{X}_i)} \right) \times (Y_i(1) - \hat{\mu}_1(\mathbf{X}_i)) \mid \mathbf{X} \right] \right] \quad (\text{LIE}) \\
&= \mathbb{E} \left[\mathbb{E} \left[\frac{T_i - \hat{e}(\mathbf{X}_i)}{\hat{e}(\mathbf{X}_i)} \mid \mathbf{X}_i \right] \times (\mathbb{E}[Y(1) \mid \mathbf{X}] - \hat{\mu}_1(\mathbf{X}_i)) \right] \quad (\text{unconfoundedness}) \\
&= \mathbb{E} \left[\left(\frac{e(\mathbf{X}_i) - \hat{e}(\mathbf{X}_i)}{\hat{e}(\mathbf{X}_i)} \right) \times (\mu_1(\mathbf{X}_i) - \hat{\mu}_1(\mathbf{X}_i)) \right].
\end{aligned}$$

This equates to zero if either $\hat{e}(\mathbf{x}) = e(\mathbf{x})$ or $\hat{\mu}_1(\mathbf{x}) = \mu_1(\mathbf{x})$. □

3.2 Asymptotic Properties

The AIPW estimator is not only robust, but also semi-parametric efficient, which very roughly translates to achieving the minimal asymptotic variance.

Theorem 5. Assume that $\hat{e}$ and $\hat{\mu}$ are consistent and jointly well estimated, i.e.

$$\|\hat{e} - e\| \times \|\hat{\mu}_t - \mu\| = o_{\mathbb{P}} \left(\frac{1}{\sqrt{n}} \right).$$

Under i.i.d. sampling, SUTVA, Unconfoundedness and Overlap we have

$$\sqrt{n}(\hat{\tau}_{\text{ATE}}^{\text{AIPW}} - \text{ATE}) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

where $\sigma^2 = \mathbb{E}[\psi_i^2]$ with

$$\psi_i = (\mu_1(\mathbf{X}_i) - \mu_0(\mathbf{X}_i)) + \left(\frac{T_i(Y_i - \mu_1(\mathbf{X}_i))}{e(\mathbf{X}_i)} - \frac{(1 - T_i)(Y_i - \mu_0(\mathbf{X}_i))}{1 - e(\mathbf{X}_i)} \right) - \text{ATE}.$$

Note that the variance can be easily estimated by

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i^2,$$

where

$$\hat{\psi} = (\hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i)) + \left(\frac{T_i(Y_i - \hat{\mu}_1(\mathbf{X}_i))}{\hat{e}(\mathbf{X}_i)} - \frac{(1 - T_i)(Y_i - \hat{\mu}_0(\mathbf{X}_i))}{1 - \hat{e}(\mathbf{X}_i)} \right) - \hat{\tau}_{\text{ATE}}^{\text{AIPW}}.$$

This only requires estimating quantities we already obtained. Note that ψ is also called the *efficient influence function*.

References

- Peng Ding. A first course in causal inference, 2023. URL <https://arxiv.org/abs/2305.18793>.
- Jianqing Fan, Kosuke Imai, Inbeom Lee, Han Liu, Yang Ning, and Xiaolin Yang. Optimal covariate balancing conditions in propensity score estimation. *Journal of Business & Economic Statistics*, 41(1):97–110, 2022.
- Jaroslav Hájek. Comment on “an essay on the logical foundations of survey sampling, part one”. In *The Foundations of Survey Sampling*, page 236. 1971.
- Daniel G Horvitz and Donovan J Thompson. A generalization of sampling without replacement from a finite universe. *Journal of the American statistical Association*, 47(260):663–685, 1952.
- Kosuke Imai and Marc Ratkovic. Covariate balancing propensity score. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 76(1):243–263, 2014.
- Guido W Imbens and Donald B Rubin. *Causal inference in statistics, social, and biomedical sciences*. Cambridge university press, 2015.
- James M Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American statistical Association*, 89(427):846–866, 1994.
- Paul R Rosenbaum and Donald B Rubin. The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55, 1983.
- Stefan Wager. Causal inference: A statistical learning approach. https://web.stanford.edu/~swager/causal_inf_book.pdf, 2024. Unpublished manuscript.